MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED: M19
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated: M19, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-413 - Visual Inspection Guide for Rubber O-Rings
MIL-STD-1167 - Ammunition Data Card

DRAWINGS

Munitions Command

F10520989 - Thruster, Cartridge Actuated, M19, Metal Parts Assembly
C8836128 - Container, Ammunition, Fiber M-445, for Thruster, Cartridge Actuated, M19 or M13 with Cartridge M119

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DRAWINGS (Cont'd)

C8836129	- Box, Packing, Ammunition for Thruster, Cartridge Actuated, M19 or M13 with Cartridge M119 in Fiber Container M445
IEL10520989	- Inspection Equipment List for Thruster, Cartridge Actuated, M19, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.3 Mechanical properties.- Mechanical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure.- The body shall withstand a constant internal pressure of 11,500 pounds per square inch (psi) minimum, for 15 seconds minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (See 6.2).

3.5 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing and the following:

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3.5.1 Torque.- Torque requirements shall be as specified in Table I. No movement or deformation shall occur upon application of the breakaway torque.

Table I

Component	A Run On (minimum)	B Including A Seating or Bottoming (inch pounds)	C Breakaway (inch pounds)
Piston and Piston Fastener	---	150 ± 20	75 min.

3.6 Protective finish.- Protective finish shall be in accordance with the applicable specification and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.7 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.8 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.8.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be made in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 5 assemblies, 10 sets of unassembled parts and fifty shear pins from the same lot of shear pins tested by the contractor. The samples shall be subjected to the tests in 4.3.3. The test procedure shall be in accordance with 4.4. The samples shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the Submission of Product provisions of MIL-STD-105.

4.3.1.2 Size of lot.- The shipping lot of metal parts assemblies shall consist of 1200 thrusters, maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

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4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects within a major or minor group the criteria in Table III shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for O-rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 -22000	300	4	5	5	6

Table III

	Total Allowable Defects Per Sample Size			
Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Sizes for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

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4.3.2.1 Pin, Shear, Firing Pin, Drawing B10520486 covering a detail of Drawing F10520989.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, max.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr or sharp corner	Visual
106. Foreign matter	Visual
MINOR	
201. Total length, max.	Gage
202. Surface finish, improper	Visual

4.3.2.2 Pin, Shear, Piston, Drawing B10520487 covering a detail of Drawing F10520989.

CRITICAL	
None defined	
MAJOR	
101. Diameter, max.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr or sharp corner	Visual
106. Foreign matter	Visual
MINOR	
201. Total length, max.	Gage
202. Surface finish, improper	Visual

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4.3.2.3 Ring, Drawing C10520489 covering a detail of Drawing F10520989.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
101. Concentricity of inside diameter with outside	Gage
102. Concentricity of pin holes	Gage
103. True position of pin holes	Gage
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Outside diameter	Gage
202. Inside diameter	Gage
203. Diameter of pin holes	Gage
204. Depth of pin holes	Gage
205. Width	Gage
206. Location of pin holes	Gage
207. Marking missing, incorrect or illegible	Visual
208. Radius or chamfer incorrect	Gage
209. Surface finish improper	Visual

4.3.2.4 Breech, Drawing D10520977 covering a detail of Drawing F10520989.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of diameter of boss with body	Gage
102. Concentricity of body with cavity	Gage
103. Concentricity of striker hole with cavity	Gage
104. Concentricity of boss cavity with cavity	Gage
105. Concentricity of groove with body	Gage
106. True position of pin hole	Gage

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR	
107. Perpendicularity of boss face with body	Gage
108. Diameter of cavity, max.	Gage
109. Diameter of body, min.	Gage
110. Diameter of groove	Gage
111. Diameter of pin hole, max.	Gage
112. Height of boss	Gage
113. Profile of groove	Gage
114. Escapement hole missing	Visual
115. Metal defective	Visual
116. Protective finish inadequate or defective	Visual
117. Surface finish of cavity improper	Visual
118. Burr or sharp corner	Visual
119. Foreign matter	Visual
MINOR	
201. Diameter of striker shoulder cavity	Gage
202. Diameter of cavity, min.	Gage
203. Diameter of body, max.	Gage
204. Diameter of pin hole, min.	Gage
205. Diameter of striker hole	Gage
206. Diameters of boss	Gage
207. Diameter of relief groove	Gage
208. Diameter of holes in relief groove	Gage
209. Depth of striker hole	Gage
210. Length from bottom of cavity to boss face	Gage
211. Total length	Gage
212. Location of basic diameter	Gage
213. Location of pin hole	Gage
214. Location of holes in relief groove	Gage
215. Diameter of escapement hole	Gage
216. Width of groove	Gage
217. Width of flange	Gage
218. Location of groove	Gage
219. Location of relief groove	Gage
220. Location of escapement hole	Gage
221. Countersink, escapement hole, missing	Visual
222. Marking missing, incorrect or illegible	Visual
223. Radius or chamfer incorrect	Gage
224. Surface finish improper, except as otherwise classified	Visual

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4.3.2.5 Pin, Firing, Drawing C10520978 covering a detail of Drawing F10520989.

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of groove with body	Gage
102.	Concentricity of striker with body	Gage
103.	Concentricity of striker shoulder with body	Gage
104.	True position of pin hole	Gage
105.	Diameter of groove	Gage
106.	Diameter of body, min.	Gage
107.	Diameter of striker	Gage
108.	Diameter of pin hole, max.	Gage
109.	Profile of striker	Gage
110.	Profile of groove	Gage
111.	Metal defective	Visual
112.	Protective finish inadequate or defective	Visual
113.	Surface finish of tip, improper	Visual
114.	Burr or sharp corner	Visual
115.	Foreign matter	Visual
MINOR		
201.	Diameter of striker shoulder	Gage
202.	Diameter of body, max.	Gage
203.	Diameter of pin hole, min.	Gage
204.	Total length	Gage
205.	Length of striker	Gage
206.	Length from body to tip of striker	Gage
207.	Location of pin hole	Gage
208.	Depth of pin hole	Gage
209.	Depth of slot in base	Gage
210.	Width of groove	Gage
211.	Width of slot in base	Gage
212.	Location of groove	Gage
213.	Marking missing, incorrect or illegible	Visual
214.	Radius or chamfer incorrect	Gage
215.	Surface finish improper, except as otherwise classified	Visual

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4.3.2.6 Cap, Base, Drawing C10520982 covering a detail of Drawing C10520975.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Minor diameter of thread, min.	Gage
103. Diameter of thread lead	Gage
104. Thread damaged or not full	Visual
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Burr or sharp corners	Visual
108. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of pilot	Gage
203. Diameter of wrench holes	Gage
204. Depth of wrench holes	Gage
205. Total length	Gage
206. Length of thread lead	Gage
207. Length of body	Gage
208. Location of wrench holes	Gage
209. Marking missing, incorrect or illegible	Visual
210. Radius or chamfer incorrect	Gage
211. Surface finish improper	Visual

4.3.2.7 Fastener, Piston, Drawing C10520983 covering a detail of Drawing C10520980.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of external thread, min.	Gage
102. Major diameter of external thread, min.	Gage
103. Pitch diameter of internal thread, max.	Gage

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MAJOR	
104. Minor diameter of internal thread, max.	Gage
105. Concentricity of pitch diameter of external thread with pitch diameter of internal thread	Gage
106. Concentricity of pilot with internal thread	Gage
107. Threads damaged or not full	Visual
108. Metal defective	Visual
109. Protective finish inadequate or defective	Visual
110. Burr	Visual
111. Foreign matter	Visual
MINOR	
201. Pitch diameter of external thread, max.	Gage
202. Pitch diameter of internal thread, min.	Gage
203. Diameter of pilot	Gage
204. Diameter of flange	Gage
205. Diameter of undercut, external	Gage
206. Diameter of undercut, internal	Gage
207. Depth to bottom of undercut, threaded cavity	Gage
208. Length of pilot	Gage
209. Total length	Gage
210. Length to flange	Gage
211. Width of undercut, external	Gage
212. Width of undercut, internal	Gage
213. Width across flats	Gage
214. Location of flats	Gage
215. Marking missing, incorrect or illegible	Visual
216. Radius or chamfer incorrect	Gage
217. Surface finish improper	Visual

4.3.2.8 Body, Drawing F10520990 covering a detail of Drawing F10520989.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of large thread, max.	Gage
102. Minor diameter of large thread, max.	Gage

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4.3.2.8 (Cont'd)

	Defect	Method of Inspection
MAJOR		
103.	Pitch diameter of thread, inlet port, max.	Gage
104.	Minor diameter of thread, inlet port, max.	Gage
105.	Pitch diameter of thread, by-pass port, max.	Gage
106.	Minor diameter of thread, by-pass port, max.	Gage
107.	Concentricity of large cavity with long cavity	Gage
108.	Concentricity of counterbore, small end, with bore	Gage
109.	Concentricity of lugs	Gage
110.	Concentricity of bore with long cavity	Gage
111.	Concentricity of grooves with bore	Gage
112.	Concentricity of pin holes	Gage
113.	True position of pin holes	Gage
114.	True position of lugs	Gage
115.	Diameter of long cavity, max.	Gage
116.	Diameter of bore, max.	Gage
117.	Diameter of lugs, min.	Gage
118.	Diameter of pin holes	Gage
119.	Diameter of cavity, inlet port	Gage
120.	Diameter of grooves	Gage
121.	Depth to bottom of large cavity, max.	Gage
122.	Depth of counterbore, inlet port	Gage
123.	Depth of counterbore, by-pass port	Gage
124.	Depth of tap, inlet port	Gage
125.	Depth of tap, by-pass port	Gage
126.	Depth to large cavity	Gage
127.	Length of thread, inlet port, max.	Gage
128.	Length of effective thread, inlet port, min.	Gage
129.	Length of effective thread, by-pass port, min.	Gage
130.	Length of thread, by-pass port, max.	Gage
131.	Location of lugs	Gage
132.	Location of inlet port	Gage
133.	Angle of chamfer, inlet port	Gage
134.	Angle of chamfer, by-pass port	Gage
135.	Angular location of hole, inlet port	Gage
136.	Diameter of chamfer, inlet port	Gage
137.	Diameter of chamfer, by-pass port	Gage
138.	Profile of grooves	Gage
139.	Internal radii forward end long cavity	Gage
140.	Radii at base of lugs	Gage
141.	Thread damaged or not full	Visual
142.	Metal defective	Visual

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MAJOR	
143. Protective finish inadequate or defective	Visual
144. Surface finish of long cavity and angles in gas ports improper	Visual
145. Burr or sharp corners	Visual
146. Foreign matter	Visual
MINOR	
201. Pitch diameter of large thread, min.	Gage
202. Pitch diameter of thread, inlet port, min.	Gage
203. Pitch diameter of thread, by-pass port, min.	Gage
204. Pitch diameter of set screw thread	Gage
205. Minor diameter of set screw thread, max.	Gage
206. Diameter of long cavity, min.	Gage
207. Diameter of large cavity	Gage
208. Diameter of lugs, max.	Gage
209. Diameter of bore, min.	Gage
210. Diameter of undercut, large thread, max.	Gage
211. Diameter of counterbore, small end	Gage
212. Diameter of counterbore, inlet port	Gage
213. Diameter of counterbore, by-pass port	Gage
214. Diameter of hole, inlet port	Gage
215. Diameter of hole, by-pass port	Gage
216. Diameter of body, small end	Gage
217. Depth to bottom of relief undercut	Gage
218. Depth to bottom of large cavity, min.	Gage
219. Depth of counterbore, small end	Gage
220. Depth of set screw thread, max.	Gage
221. Depth to long cavity	Gage
222. Length to intersection of face, inlet port, and head diameter	Gage
223. Total length	Gage
224. Length of lugs	Gage
225. Length of effective set screw thread, min.	Gage
226. Length of head	Gage
227. Length from lugs to face at small end	Gage
228. Length of by-pass port	Gage
229. Width across head, at lug	Gage
230. Width of relief undercut	Gage

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MINOR	
231. Location of hole, inlet port	Gage
232. Location of pin holes	Gage
233. Location of grooves	Gage
234. Location of by-pass port	Gage
235. Centrality of head	Gage
236. Height of by-pass port	Gage
237. Diameter of undercut, long cavity, max., if applicable	Gage
238. Depth of cavity, inlet port	Gage
239. Width of undercut, long cavity, if applicable	Gage
240. Width of grooves	Gage
241. Width of undercut at large cavity	Gage
242. Spot face improper	Visual
243. Marking missing, incorrect or illegible	Visual
244. Radius or chamfer incorrect, except as otherwise classified	Gage
245. Surface finish improper, except as otherwise classified	Visual

4.3.2.9 Piston, Drawing D10520992 covering a detail of Drawing C10520991.

CRITICAL

None defined

MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of thread with body	Gage
104. Concentricity of shoulder with body	Gage
105. Concentricity of head with body	Gage
106. Concentricity of groove with head	Gage
107. Concentricity of cavity with body	Gage
108. Diameter of head, min.	Gage
109. Diameter of body, min.	Gage
110. Diameter of shoulder, min.	Gage
111. Diameter of groove	Gage
112. Profile of groove	Gage
113. Thread damaged or not full	Visual
114. Metal defective	Visual

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4.3.2.9 (Cont'd)

Defect	Method of Inspection
MAJOR	
115. Protective finish inadequate or defective	Visual
116. Surface finish of body, improper	Visual
117. Burr or sharp corners	Visual
118. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of head, max.	Gage
203. Diameter of body, max.	Gage
204. Diameter of shoulder, max.	Gage
205. Diameter of undercut, body, if applicable	Gage
206. Diameter of cavity	Gage
207. Diameter of undercut, at thread, min.	Gage
208. Length to body	Gage
209. Total length	Gage
210. Length of shoulder	Gage
211. Location of head	Gage
212. Location of groove	Gage
213. Location of holes in body	Gage
214. Diameter of escapement hole in head	Gage
215. Diameter of holes in body	Gage
216. Depth of cavity	Gage
217. Width of groove	Gage
218. Width of undercut, body, if applicable	Gage
219. Width of thread undercut	Gage
220. Marking missing, incorrect or illegible	Visual
221. Radius or chamfer incorrect	Gage
222. Surface finish improper except as otherwise classified	Visual

4.3.2.10 Base Cap and Pellet Assembly, Drawing C10520975, covering a detail of Drawing F10520989.

CRITICAL

None defined

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4.3.2.10 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Protrusion of pellets	Gage
102. Pellet missing, loose or improperly inserted	Manual & Visual
103. Thread damaged	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Location of pellets	Gage

4.3.2.11 Piston and Pellet Assembly, Drawing C10520991, covering a detail of Drawing F10520989.

CRITICAL	
None defined	
MAJOR	
101. Protrusion of pellet	Gage
102. Pellet missing, loose or improperly inserted	Manual & Visual
103. Thread damaged	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Location of pellet	Gage

4.3.2.12 Piston Fastener and Pellet Assembly, Drawing C10520980 covering a detail of Drawing F10520989.

CRITICAL	
None defined	
MAJOR	
101. Protrusion of pellets	Gage
102. Pellet missing, loose or improperly inserted	Manual & Visual

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4.3.2.12 (Cont'd)

Defect	Method of Inspection
MAJOR	
103. Thread damaged	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Location of pellets	Gage

4.3.2.13 Breech and Firing Pin, Subassembly, consists of: Breech 10520977, Firing Pin 10520978 and O-ring MS29513-9, covering a detail of Drawing F10520989.

CRITICAL	
1. Firing pin protrusion	Gage
MAJOR	
101. Protective finish inadequate or defective	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.3.2.14 Metal Parts Assembly, Drawing F10520989.

CRITICAL	
None defined	
MAJOR	
101. Any part missing, except as otherwise classified	Visual
102. Improper assembly	Visual
103. Thread damaged or not full	Visual
104. Foreign matter	Visual

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4.3.2.14 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Protective finish inadequate or defective	Visual
202. Barrier material missing from set screws	Visual
203. Shipping cap or plug missing	Visual
204. Pellet missing from piston fastener	Visual

4.3.2.15 Fiber Container, Drawing C8836128.

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

4.3.2.16 Box, Packing, Drawing C8836129.

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual & Manual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual

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4.3.2.16 (Cont'd)

Defect	Method of Inspection
MINOR	
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness.- Fifty units from each heat treated lot of components shall be tested for compliance with the values noted on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.2 Hydrostatic pressure.- Each body shall be subjected to an internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.4 shall be cause for rejection of the unit.

4.3.3.3 Shear pin strength.- Fifty pins of each type from each lot shall be tested for shear strength. Failure of any shear pin to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.4 Breakaway torque.- Each piston fastener and piston shall be subjected to a breakaway torque. Failure of any unit to comply with the requirements of column C in 3.5.1 shall be cause for rejection of the unit.

4.3.3.5 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable MIL-Standard. Failure of any sample to comply shall be cause for rejection of the lot.

4.3.3.6 Residual magnetism.- Each assembly shall be subjected to a test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.7 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10520989.

4.4 Test methods and procedure.

4.4.1 Hardness.- Prior to the application of protective finish, a hardness tester shall be used to determine the values specified on the detail drawing.

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4.4.2 Hydrostatic pressure.- Prior to the application of protective finish, the body shall be mounted in a device and subjected to the internal hydrostatic pressure in accordance with the requirements of 3.4. The testing device used shall internally seal the body at each end without applying any axial load other than that applied by the liquid pressure.

4.4.3 Shear pin strength.- After application of protective finish, the pins shall be subjected to the shear strength test. The load shall be applied at the steady rate of 0.125 inches per minute.

4.4.4 Breakaway torque.- A torque wrench set at the torque requirement specified in Table I, in 3.5.1, shall be used in conjunction with a fixture to determine the adequacy of the joint. The piston shall be secured in an air vise or other fixture. The torque shall be applied to the piston fastener.

4.4.5 Protective finish.- Test procedures for protective finish shall be in accordance with the MIL-Standard referenced on the component drawing.

4.4.6 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as the compass, in a north-south horizontal position with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with Drawings C8836128 and C8836129.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A264

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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